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Exercise Sheet 2

Exercise 1 — Bond lengths and angles in bcc and fcc [moderate]

The symmetric primitive vectors for the body-centered cubic (bcc) lattice are

$$\vec{a}_1 = \frac{a}{2}(-1, 1, 1), \quad \vec{a}_2 = \frac{a}{2}(1, -1, 1), \quad \vec{a}_3 = \frac{a}{2}(1, 1, -1).$$

For the face-centered cubic (fcc) lattice they are

$$\vec{a}_1 = \frac{a}{2}(0, 1, 1), \quad \vec{a}_2 = \frac{a}{2}(1, 0, 1), \quad \vec{a}_3 = \frac{a}{2}(1, 1, 0).$$

Calculate, for each case:

- the lengths of the primitive vectors $|\vec{a}_i|$ in terms of the cubic lattice constant a ,
- the angles between the primitive vectors (i.e. $\angle(\vec{a}_i, \vec{a}_j)$ for $i \neq j$).

Exercise 2 — Symmetry operations of simple objects [advanced]

For each of the following objects, list all **symmetry operations** that leave it invariant. Indicate the number and type of *rotation axes*, *mirror planes*, and whether an *inversion center* exists.

- A line segment along the z -axis, extending symmetrically about the origin.
- A regular hexagon lying in the xy -plane, centered at the origin.
- A rectangular cuboid (box) with edge lengths $a \neq b \neq c$, centered at the origin and aligned with the Cartesian axes.
- A tetrahedron with all edges equal.

Exercise 3 — Simple cubic lattice dressed with pentagons [moderate]

Consider a simple cubic Bravais lattice. At each lattice site sits a regular pentagon (a molecule with a fivefold symmetry axis). All pentagons share the same orientation:

- the fivefold axis is along the y -axis,
- the pentagon lies parallel to the xz -plane,
- one corner points along the z -axis.

Which **point group** describes the symmetry of this crystal?



Exercise 4 — *Different crystal planes in a cubic lattice*

[moderate]

Consider a crystal with a simple cubic Bravais lattice with lattice constant a .

- (a) Determine the Miller indices (hkl) of the planes that cut the axes at
- (i) $x = a, y = a, z = a/2$;
 - (ii) $x = a, y = \infty, z = a$;
 - (iii) Is it possible to specify the Miller indices for a plane that cuts the axes at $x = \infty, y = \infty, z = a/2$? Explain your answer.
- (b) For the following sets of planes, state whether they are equivalent by symmetry in a cubic crystal:
- (i)
 $(100), (010), (001);$
 - (ii)
 $(110), (101), (011);$
 - (iii)
 $(111), (1\bar{1}1), (\bar{1}11), (11\bar{1}).$
- (c) Using the cubic formula
- $$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}},$$
- calculate and compare the interplanar spacings d_{100} , d_{110} , and d_{111} for $a = 4.0 \text{ \AA}$.
- (d) Which family of crystal planes $\{hkl\}$ in an fcc crystal has the *highest planar density* of atoms?
- (e) Sketch (or describe) how the (100), (110), and (111) planes intersect the simple cubic cell and discuss how their atomic arrangements differ on the plane surface.